SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Answer Script Evaluation Sheet

Continuous Internal Assessment 1/2

Date:

Programme	B.E/B.Tech	**Slot**	C/D
Course Code	CSA0709/ CSA0710	**Course Name**	Computer Networks
Duration	1.30 Hrs	**Marks**	50

Q.No.	Max. Marks	Awarded Marks
1.	10	10
2.	10	8
3.	10	9
4.	10	9
5.	10	7
Total	50	43

Course Faculty: Dr. Shankar R.,

Computer networks

Name :- Vedala Jeevan
Reg no :- 192526013 3

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CIA-1

1) Data Link Layer

The Data Link Layer is responsible for a transfering data between two directly connected devices over a Physical Network.

Functions

1) framing :- Divides data into frames for transmi identify devices on a local network.

2) Physical addressing :- Uses mac addresses to identify devices on a local network.

3) Error Detection :- detects transimission errors using methods such as CRC.

4) flow control :- Helps prevent a sender from overwhelming a receiver when supported by the link protocol

Transport Layer

The Transport Layer provides communication between applications running on different devices.

- Segmentation : Divides application data into

smaller segments.

- Error recovery: TCP detects missing data.

b)

Feature	Connection-oriented service	Connectionless service.
Protocol	TCP	UDP
Connected Setup	Required	Not required
Reliability	supports reliable delivery	No delivery guarantee
Acknowledgement	uses acknowledgements	No built in acknowledgments
Data ordering	maintains ordered delivery	No ordering guarantee
Speed and Overhead	Generally more overhead	Generally lower overhead
Example	file transfer, web browsing using HTTPS over TCP	online gaming DNS queries live streaming UDP.

TCP is preferred when reliable ordered delivery is required, whereas UDP is useful important

2)

a) Encapsulation

It is a process of adding protocol information to data as it moves down the OSI layers of the source device.

OSI Layer	Process	Data Limit
7. Application	Creates application data	Data
6. Presentation	Handles translation, encryption or the compression as needed	Data
5. Session	manages communication sessions	Data
4. Transport	Adds transport header such as TCP or UDP	Segment
3. Network	Adds MAC addressing and trans checking information	Packet.
2. Data Link	Address source	frame
1. Physical	Transmits bits over the medium	Bits

3) Classful IP Addressing

In classful addressing, IPv4 addresses were divided into fixed classes A, B, C, D and E

Class	First octet Range	Default Mask	Purpose
A	1-126	255.00 (/8)	Large networks
B	128 - 191	255.255.00 (/16)	Medium Sized
C	192-223	255.255.0 (/24)	Small networks
D	224-239	Not applicable	Multicas
E	240 -255	Not applicable	Reserved.

CIDR improves IPv4 address utilization and routing efficiency by using flexible network Prefix lenghts. Smaller organizations can receive smaller address blocks.

4) Given

- Network address: 172.16.0.0/16
- Required subnets: At least 6

Step 1 :-

$2^n \geq$ Required subnets

for 6 subnets

$2^2 = 4$

$2^3 = 8$

∴ 3 bits must be borrowed

Step 2

Prefix = /16

Borrowed = 3

New Prefix = 16+3 = 19

∴ new CIDR Prefix is /19 255.5 = 255.224

∴ CIDR notation. /19

Dotted decimal Subnet mask : 255.255.224.0

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5) Introduction

Routing is the process of selecting the best Path for sending data Packets from a source

network to a destination network. Routers use routing tables to decide where to forward packets.

There are two main types of routing

1) Static Routing
2) Dynamic Routing

Both routing methods can be configured and tested using network simulation tools such as Cisco Packet Tracer.

1) Static Routing

Static routing is a routing method in which the network administrator manually enters routes into the router's routing table.

* Routes are configured manually by the administrator
* It does not automatically change when the network topology changes
* It is suitable for small networks.
* It provides better control over the routing path

Example :- A static route can be configured using the following command.